lit2026 W8: summation-polynomial model checks

2026-10-11 · branch `lit2026/w8-summation-models` (stacked on aburan28/crypto#1681) · program baseline `4b6e09b78`

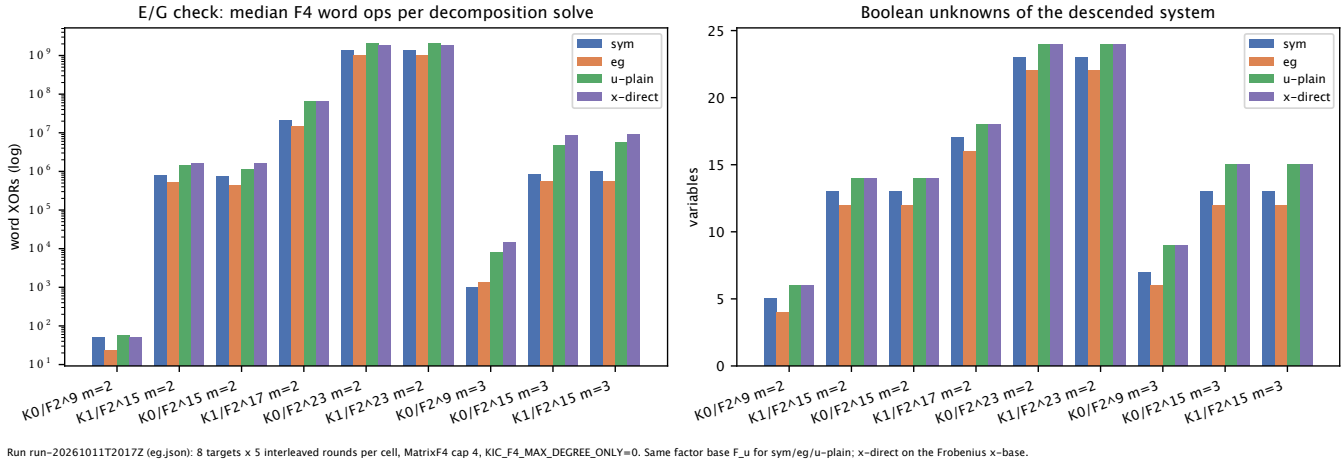
Question. Do four 2026 papers on curve models and system modelling (Dryło ePrint 2026/1299; Wu arXiv 2608.01675; Bariant ePrint 2026/1927; La Scala–Marchesin–Tiwari arXiv 2609.18866) change the cost of the repository's Weil-descended decomposition systems over GF(2^n)? All code is Rust (`examples/lit2026_w8_summation_models.rs`) on the unchanged library; systems are solved by the repo's Boolean matrix-F4 with splitting, all roots enumerated; arms interleaved, 8 targets x 5 rounds per cell; deterministic gate = F4 word XORs; wall time L0 on a host at load 80–360. Run reports: `reports/run-20261011T2017Z/`, `reports/run-20261011T2131Z/`. Full tables: `README.md`.

1. E/G check (Dryło): symmetrisation = E/⟨T⟩, plus a cover at m = 3

On a Koblitz curve the FGHR invariant is $w = u^2 + u = 1/X \circ \Phi$ with $X = x + \sqrt{b}/x$ the Vélu abscissa of $\Phi : E \to E/\langle \mathrm{T} \rangle$, and $b'' = 1$ (checked on 33 relations per cell). The repo's symmetrised polynomial is $G(w) + sH(w)$ with $s = \sum u$, $s^2 + s = \sum w$, and

$$\mathrm{Res}_s\Big(G + sH,\, s^2 + s + \sum w\Big) = G^2 + GH + H^2 \sum w = S_{m+1}^{E/\langle \mathrm{T} \rangle}(1/w) \cdot \prod w^{2^{m-1}} \quad \text{over } \mathbb{F}_2,$$

certified as **IDC1hc7b594938b50f160** ($m = 2$) and **IDC1hcf83176ae9a93715** ($m = 3$). At $m = 2$, $H = 1$: the repo's system is the $E/\langle \mathrm{T} \rangle$ system plus a linearly eliminable parity bit. At $m = 3$, s multiplies H: the repo's system is the $E/\langle \mathrm{T} \rangle$ S_4 (Boolean degree 6) plus an auxiliary cover that lowers the degree to 4. Root sets (as w-tuples) were identical across `sym`, `eg` and `u-plain` on all 72 targets.

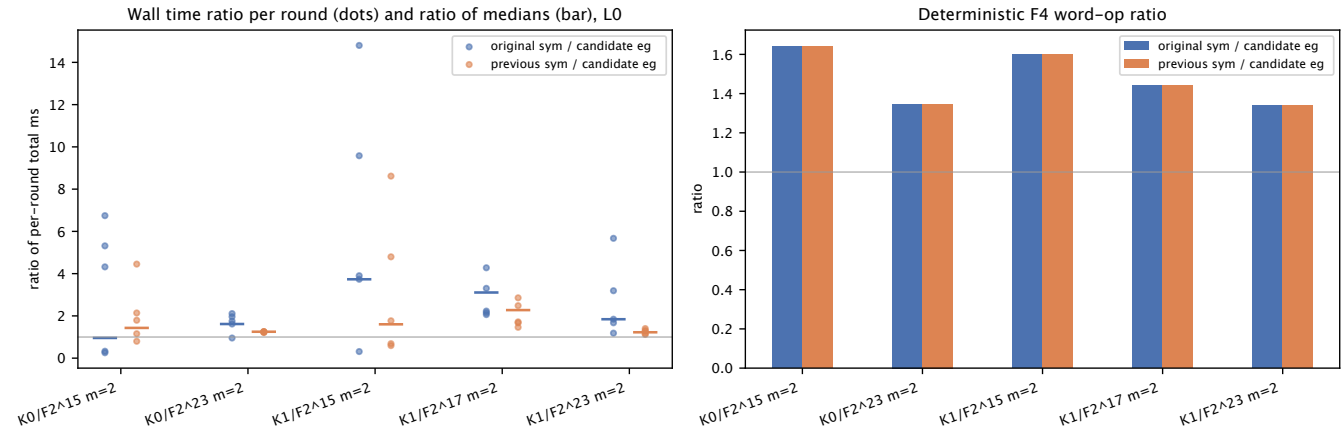


Run run-20261011T2017Z (eg.json): 8 targets x 5 interleaved rounds per cell, MatrixF4 cap 4, KIC_F4_MAX_DEGREE_ONLY=0. Same factor base F_u for sym/eg/u-plain; x-direct on the Frobenius x-base.

Figure 1: Median word XORs per solve and Boolean unknowns, per arm and cell (run-20261011T2017Z).

At K1/F2^15, m = 3 (median ms; word ops): x-direct 166.7; 9.01M ⟶ u-plain 64.5; 5.77M ⟶ eg 20.7; 0.54M ⟶ sym 13.1; 1.02M. Interpretation: most of the repo's "symmetry" gain is the change of curve to $E/\langle \mathrm{T} \rangle$ that the invariant coordinate is; the non-relabelling part is the s-cover at $m = 3$ (1.6x wall time, 1.9x more word ops, fewer splits).

2. The faster variant: solve E/⟨T⟩ directly at m = 2



Run run-20261011T2131Z (compare.json): original = harness on the 4b6e09b78 library, sym arm; candidate binary runs sym (previous) and eg in-process; 5 rounds ABAB, 8 targets per cell.

Figure 2: Interleaved original (4b6e09b78 library, `sym`) / previous (`sym`) / candidate (`eg`), 5 rounds ABAB, 8 targets per cell (run-20261011T2131Z).

cell (ICV1)	word ops orig = prev	cand	ratio	wall prev/cand, median [min–max]
K1/F2^15 `icv1-f2m15-t275-b7f03703`	6 750 384	4 218 414	1.600	1.60 [0.59–8.62]
K0/F2^15 `icv1-f2m15-tm275-2d22ff5d`	6 235 447	3 797 654	1.642	1.43 [0.80–4.45]
K1/F2^17 `icv1-f2m17-tm101-00378d4e`	169 687 396	117 819 615	1.440	2.27 [1.46–2.86]

K0/F2^23 `icv1-f2m23-t5197-69e76b73`	11 054 138 493	8 205 050 470	1.347	1.25 [1.21–1.26]
K1/F2^23 `icv1-f2m23-tm5197-1f85e9e1`	10 866 191 751	8 107 487 123	1.340	1.22 [1.12–1.40]

Word-op ratios are deterministic (identical in all rounds); root digests identical across the three arms. Original/candidate wall ratios are in the run report; original and previous run the same code and their cross-process spread (up to 5.4x) is the noise floor. Scope: the m = 2 solve on toy cells; lifting an `eg` root tries both parities.

3. Wu's binary state model, Frobenius auxiliary, convolution splitting

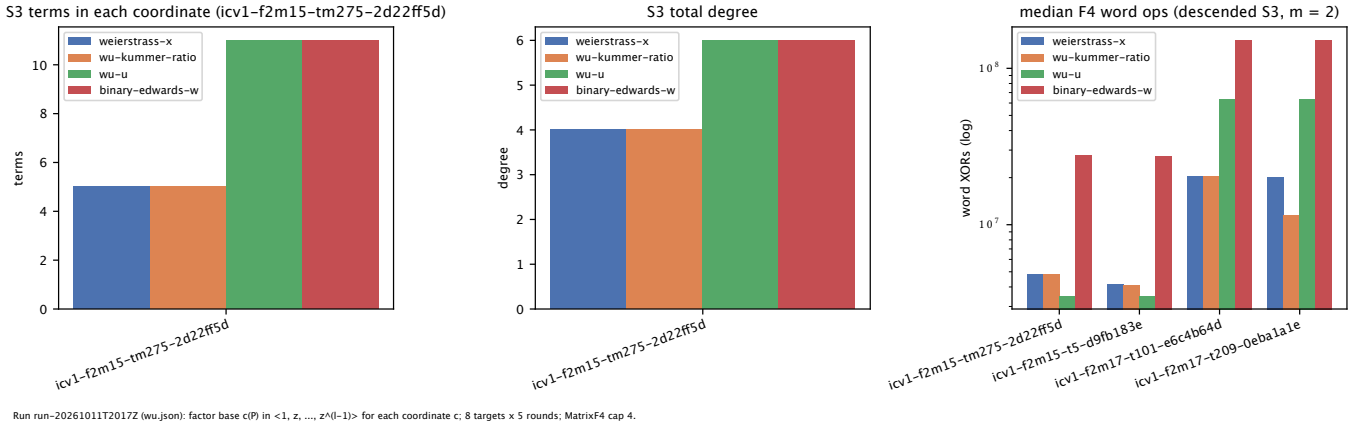


Figure 3: S3 per coordinate: Weierstrass x, Wu Kummer ratio k = x/d, Wu u = d/(x+d) (the FGHR frame), binary Edwards w (an E/⟨T⟩ coordinate).

Wu's u is a Möbius frame of x and is the FGHR frame ($u \to u + 1$ is translation by T; $d = b^{1/4}$; the model needs $\mathrm{Tr}(a) = 0$). Its S_3 has 11 terms and total degree 6 against 5 and 4; descended monomials 66–83 against 81–100; word ops 1.2–1.4x fewer at n = 15 and 3.1x more at n = 17. Binary Edwards w is invariant under T (a Dryło compression) and costs 5.7–7.5x more word ops on these bases. No model is uniformly cheaper.

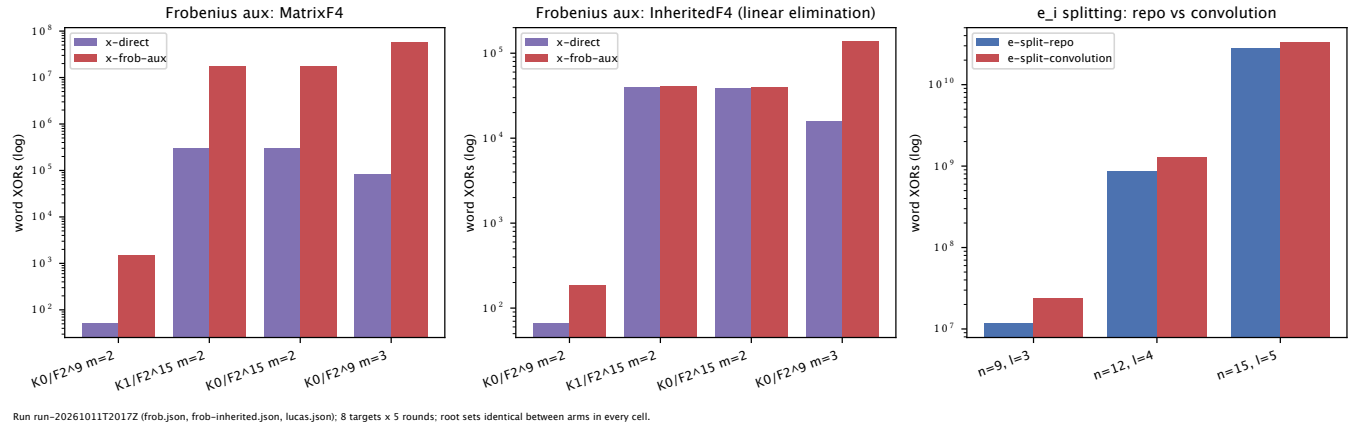


Figure 4: Frobenius auxiliary blocks (MatrixF4 and InheritedF4) and e_i splittings.

Frobenius auxiliary (negative). Over the F2 descent $x \to x^{2^k}$ is linear, so the auxiliary is in the span of the unknowns: same Boolean degree, same ideal after linear elimination. MatrixF4: 57–58x (m = 2) and 701x (m = 3) more word ops; InheritedF4 (eliminates linear generators): 1.03x (m = 2), 8.6x (m = 3).

Convolution splitting (negative). Degree 3 ⟶ 2 and 4–14% fewer monomials, the same split tree, 1.21–2.07x more word ops and 2.4–2.9x the wall time at n = 9, 12, 15. Lucas' factorisation applies to symmetric functions of F2-valued variables and has no counterpart here; m = 4 is measured only at correspondence level (no binary symmetrised S5 in the repo).

4. Status and limits

E/G check (2-torsion, m = 2, 3): complete except the K1/F2^17 m = 3 cell, stopped unfinished after 2 h. 4-torsion: no 4-torsion-symmetrised binary system exists in the repo. Wu: complete. Frobenius auxiliary: complete (negative). Convolution splitting: partial (S4 full; m = 4 correspondence only). Speedup: `eg` at m = 2, 1.34–1.64x word ops, interleaved with original and previous. Toy fields n ≤ 23 only; no IC-versus-rho, exponent or deployed-curve statement. Found: the F4 engine's wide-system path switches at 24 variables, leaving the 22–23-variable `sym`/`eg` systems at n = 23 on a path roughly 4x slower in a cross-session observation (README).